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COLORADO SCHOOL OF MINES
MATH 455: Partial Differential Equations

Assignment #6 - Problems in Higher Dimensions
Due Tuesday, November 17, 2026

1. (12 points) Consider the doubly-indexed Fourier series

$$f(x,y) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} b_{mn} \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right)$$

for a given function $f \in \mathrm{PC}^1\left([0,a]\times[0,b]\right)$.

(a) Rewrite the series in the form

$$f(x,y) = \sum_{m=1}^{\infty} B_m(x) \sin\left(\frac{m\pi y}{b}\right) \tag{1}$$

and identify $B_m(x)$ in terms of the expressions given above.

Solution —

$$\begin{aligned} f(x,y) &= \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} b_{mn} \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right) \\ &= \sum_{m=1}^{\infty} \sin\left(\frac{m\pi y}{b}\right) \underbrace{\left[\sum_{n=1}^{\infty} b_{mn} \sin\left(\frac{n\pi x}{a}\right)\right]}_{B_m(x)} \end{aligned}$$

$$\boxed{B_m(x) = \sum_{n=1}^{\infty} b_{mn} \sin\left(\frac{n\pi x}{a}\right)}$$

(b) Consider $0 < x < a$ fixed. Then (1) is a Fourier series in y. Using our previous knowledge of (singly-indexed) Fourier series, write down an expression for $B_m(x)$ in terms of $f(x,y)$ and $\sin\left(\frac{m\pi y}{b}\right)$.

Solution — If we fix $x = x_0$, then $B_m(x) = B_m(x_0)$ becomes a function only of m. Let us call this function $C_m = B_m(x_0)$. So

$$\begin{aligned} f(x_0,y) &= \sum_{m=1}^{\infty} \sin\left(\frac{m\pi y}{b}\right) B_m(x_0) \\ &= \sum_{m=1}^{\infty} C_m \sin\left(\frac{m\pi y}{b}\right) \end{aligned}$$

Since this is the Fourier sin-expansion of $f(x_0, y)$ on $y \in [0, b]$, then the Fourier coefficient C_m must be

$$C_m = \frac{2}{b} \int_0^b f(x_0, y) \sin\left(\frac{m\pi y}{b}\right) dy$$

So

$$\boxed{B_m(x) = \frac{2}{b} \int_0^b f(x, y) \sin\left(\frac{m\pi y}{b}\right) dy}$$

(c) Consider $m \in \mathbb{N}$ fixed. Using the expression for $B_m(x)$ derived in part (a) and our previous knowledge of Fourier series, solve for b_{mn} in terms of $B_m(x)$ and $\sin\left(\frac{n\pi x}{a}\right)$.

Solution — Fix $m = m_0$. Then from

$$B_m(x) = \sum_{n=1}^{\infty} b_{mn} \sin\left(\frac{n\pi x}{a}\right)$$

$$\implies B_{m_0}(x) = \sum_{n=1}^{\infty} b_{m_0 n} \sin\left(\frac{n\pi x}{a}\right)$$

Hence

$$B_{m_0 n} = \frac{2}{a} \int_0^a B_{m_0}(x) \sin\left(\frac{n\pi x}{a}\right) dx$$

And, re-generalizing to to non-fixed m,

$$\boxed{b_{mn} = \frac{2}{a} \int_0^a B_m(x) \sin\left(\frac{n\pi x}{a}\right) dx}$$

(d) Finally, substitute the formula for B_m derived in part (b) within the formula for b_{mn} found by part (c), and use this to identify a final formula for the Fourier coefficients b_{mn} that depends upon $f(x, y)$.

Solution —

$$\begin{aligned} b_{mn} &= \frac{2}{a} \int_0^a B_m(x) \sin\left(\frac{n\pi x}{a}\right) dx \\ &= \frac{2}{a} \int_0^a \left[\frac{2}{b} \int_0^b f(x, y) \sin\left(\frac{m\pi y}{b}\right) dy\right] \sin\left(\frac{n\pi x}{a}\right) dx \\ &= \boxed{\frac{4}{ab} \int_0^a \int_0^b f(x, y) \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right) dy dx} \end{aligned}$$

2. (18 points) Find the solution $u(t, x, y)$ of the wave equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} - \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = 0 \qquad \text{(WE)}$$

on the square

$$\Omega = \{(x, y) : 0 < x < \pi, 0 < y < \pi\}$$

with homogenous Dirichlet boundary conditions and the initial conditions

$$u(0, x, y) = xy(\pi - x)(\pi - y) \qquad \text{and} \qquad \frac{\partial u}{\partial t}(0, x, y) = 0$$

for $(x, y) \in \Omega$. As in class, you may assume that the solution is oscillatory in t. Also, notice the symmetry in $u(0, x, y)$. Upon separating variables, you do NOT need to go through all of the cases to solve for the eigenfunctions and eigenvalues. Just use our previous work!